

HAT-P-36b

R-1325

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210204 · created 2026-07-31T20:05:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459249.9025 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.045
Transit depth (Rp/Rs)^2	2.16 +/- 1.32 [%]
Orbital Inclination (inc)	80.31 +/- 2.88
Airmass coefficient 1 (a1)	0.026 +/- 0.014
Airmass coefficient 2 (a2)	2.95 +/- 0.22
Scatter in the residuals of the lightcurve fit is	31.82 %
Best Comparison Star	#1 - [606, 340]
Optimal Aperture	1.87
Optimal Annulus	11.0
Transit Duration (day)	0.071 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.147 $\pm$ 0.045 vs 0.126 (deviation 0.3 σ)
Duration fit vs published	0.071 d vs 0.0925 d (deviation 0.66 σ)
Mid-time O-C	8.4 min
Depth SNR	2.1
Residual scatter	31.82 %

Light curve

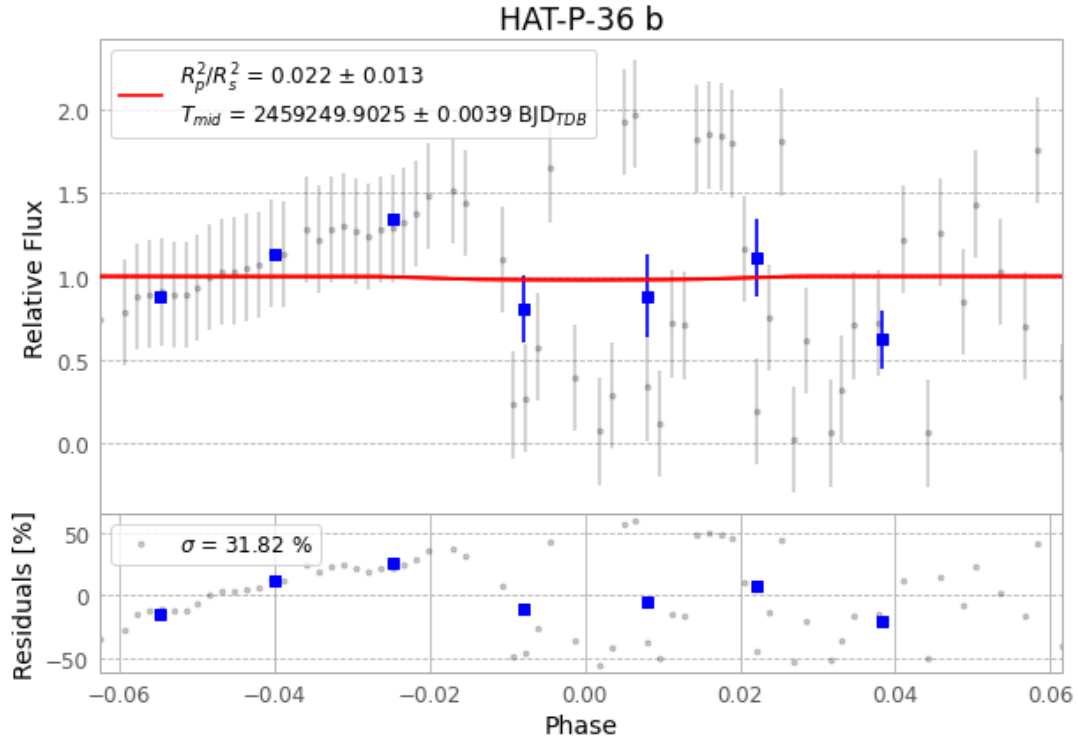


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-02-04.png (SHA-256 44997a537e0d5056...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [613, 356], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ca95a065bf1e591b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), HATP-36210204073310.FITS → HATP-36210204113012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210204.log (886e22967427...), FinalLightCurve_HAT-P-36 b_2021-02-04.png (44997a537e0d...), FinalLightCurve_HAT-P-36 b_2021-02-04.csv (02454acc2785...)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:05Z.